

Revision

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$$P = -\frac{2}{3} \frac{q^2}{m^2 c^3} \left[\frac{dp_\mu}{d\tau} \frac{dp^\mu}{d\tau} \right]_{ret} \quad (1)$$

$$p^\mu = \left(\frac{E}{c}, \vec{p} \right), \quad E = \gamma m c^2, \quad \vec{p} = \gamma m c \vec{\beta} \quad (2)$$

$$d\tau = \frac{dt}{\gamma} \quad (3)$$

$$\frac{dp^\mu}{d\tau} = \frac{dt}{d\tau} \frac{dp^\mu}{dt} = \gamma \frac{d}{dt} \left[\gamma m c (1, \vec{\beta}) \right] = m c \gamma (\dot{\gamma}, \dot{\gamma} \vec{\beta} + \gamma \dot{\vec{\beta}}) \quad (4)$$

$$\left(\vec{\beta} \times \dot{\vec{\beta}} \right)^2 = \vec{\beta}^2 \dot{\vec{\beta}}^2 - \left(\vec{\beta} \cdot \dot{\vec{\beta}} \right)^2 \quad (5)$$

(b)

$$P = \frac{2}{3} \frac{q^2}{c^7} \frac{E^4 \beta^4}{m^4 \rho^2} \quad (6)$$

$$r(t) = \rho (\cos(\omega t), \sin(\omega t), 0)^T \quad (7)$$

$$cb = \dot{r} = \omega \rho (-\sin(\omega t), \cos(\omega t), 0)^T \quad (8)$$

$$c\dot{\beta} = \ddot{r} = \omega^2 \rho (-\cos(\omega t), -\sin(\omega t), 0)^T = -\omega^2 r(t) \quad (9)$$

The acceleration and velocity are orthogonal so the dot product vanishes and cross product.